



LAB # 06 — TASKS

Implementation of Telnet & SSH in CPT

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Section	BAI-5A	Date of Submission	

Submission Guidelines

- Complete all tasks and include screenshots of every configuration in this Word file.
- Upload all tasks along with the **.pkt** files to your Google Drive and make sure to grant access to abdullah.shaikh@nu.edu.pk.
- Submit the **Google Drive link** by pasting it into a **Word/PDF** document on **GCR** within the deadline.
- Name the file as **LabCourseName_Lab#-YourName-YourRollNo/ID** — e.g. **CN_Lab#06-Abdullah Shaikh-25K-7826**.
- *No deadline will be extended under any circumstances.*

OBJECTIVE

To configure remote management of network devices in Cisco Packet Tracer — enabling Telnet access on switches and replacing it with secure SSH access on a router — and to understand why SSH is preferred over Telnet for remote administration.

TASK # 01 — REMOTE SWITCH MANAGEMENT USING TELNET

You are a **network administrator** tasked with configuring remote access for network switches in an office environment. The company wants to ensure that IT personnel can manage switches remotely via **Telnet** from the nearest PCs connected to them.

Your objective is to:

1. **Enable Telnet access** on three switches (S1, S2, S3).
2. **Assign IP addresses** to the switches for remote access.
3. **Ensure that the nearest PC to each switch can successfully establish a Telnet session.**
4. **Verify connectivity** using Telnet and relevant network commands.

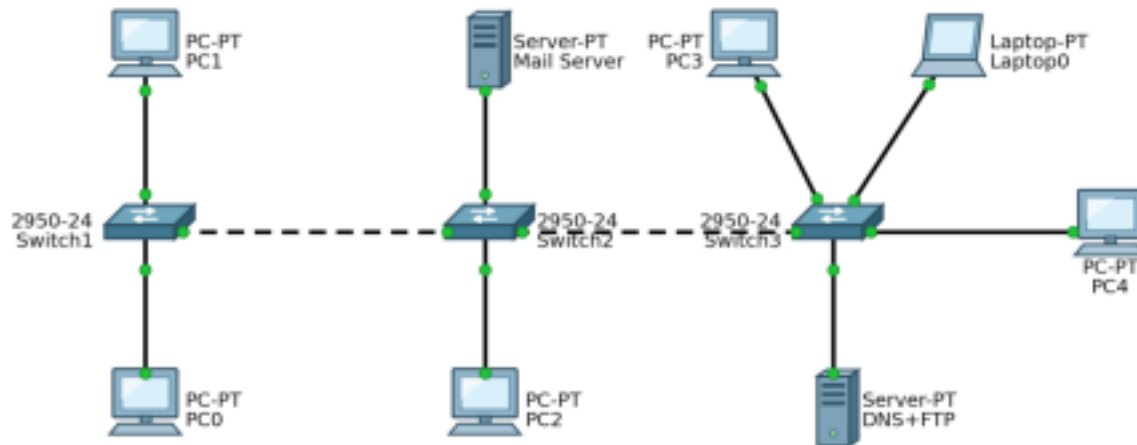


Figure 1:

Topology for Task #01

Note

Assign IP addresses to the computers. The network should follow this format: **XX.XX.YY.0**, where **XX.XX** is derived from your roll number. For example, if your roll number is **3879**, the network should be **38.79.1.0**. For all other networks, **Y** should be replaced sequentially (e.g., **2, 3, etc.**).

PASTE YOUR TELNET CONFIGURATION AND VERIFICATION SCREENSHOTS HERE

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#enable password cisco
SW1(config)#line vty 0 4
SW1(config-line)#password cisco
SW1(config-line)#login
SW1(config-line)#transport input telnet
SW1(config-line)#exit
SW1(config)#interface input telnet
      ^
% Invalid input detected at '^' marker.

SW1(config)#interface vlan 1
SW1(config-if)#ip address 24.13.1.11 255.255.255.0
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#write memory
Building configuration...
[OK]
SW1#
```

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IOS Command Line Interface

```
% Invalid input detected at '^' marker.  
Switch(config)#configure terminal  
^  
% Invalid input detected at '^' marker.  
Switch(config)#configure terminal  
^  
% Invalid input detected at '^' marker.  
Switch(config)#hostname SW2  
SW2(config)#enable password cisco  
SW2(config)#line vty 0 4  
SW2(config-line)#password cisco  
SW2(config-line)#login  
SW2(config-line)#transport input telnet  
SW2(config-line)#exit  
SW2(config)#interface vlan 1  
SW2(config-if)#ip address 241.13.1.12 255.255.255.0  
Not a valid host address - 241.13.1.12  
SW2(config-if)#no shutdown  
  
SW2(config-if)#  
%LINK-3-UPDOWN: Interface Vlan1, changed state to down  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up  
exit  
SW2(config)#end  
SW2#  
%SYS-5-CONFIG_I: Configured from console by console  
write memory  
Building configuration...  
[OK]  
SW2#  
SW2#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
SW2(config)#interface vlan 1  
SW2(config-if)#ip address 24.13.1.12 255.255.255.0  
SW2(config-if)#no shutdown  
SW2(config-if)#exit  
SW2(config)#exit  
SW2#  
%SYS-5-CONFIG_I: Configured from console by console  
write memory  
Building configuration...  
[OK]  
SW2#
```

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```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#enable password cisco
SW3(config)#line vty 0 4
SW3(config-line)#password cisco
SW3(config-line)#login
SW3(config-line)#transport input telnet
SW3(config-line)#exit
SW3(config)#interface vlan 1
SW3(config-if)#ip address 24.13.1.13 255.255.255.0
SW3(config-if)#no shutdown

SW3(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
SW3(config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
SW3#

```

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PC 0:

```

Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 24.13.1.11
Trying 24.13.1.11 ...
% Connection timed out; remote host not responding
C:\>clear
Invalid Command.

C:\>
C:\>telnet 24.13.1.11
Trying 24.13.1.11 ...Open

User Access Verification

Password:
Password:
SW1>enable
Password:
SW1#

```

Pc2:

```

C:\>telnet 24.13.1.12
Trying 24.13.1.12 ...Open

User Access Verification

Password:
SW2>enable
Password:
SW2#

```

PC3:

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 24.13.1.13
Trying 24.13.1.13 ...Open

User Access Verification

Password:
SW3>enable
Password:
SW3#
```

TASK # 02 — SECURING ROUTER ACCESS WITH SSH

Your company's **network security team** has detected a **data leak** caused by the use of **Telnet**, which transmits data in plain text. To prevent further security risks, the IT team has decided to **disable Telnet and enable SSH (Secure Shell) for remote access** on all network devices, starting with the **main router (R1)**.

Currently, one **interface of the router (Fa0/0)** is correctly configured, but the **second interface (Fa0/1)** remains unconfigured.

Your task is to:

1. **Enable SSH on Router R1** to allow secure remote administration.
2. **Verify that you can successfully connect to the router using SSH** from a designated PC0.
3. **Once connected, configure the second interface (Fa0/1) with the appropriate IP settings.**

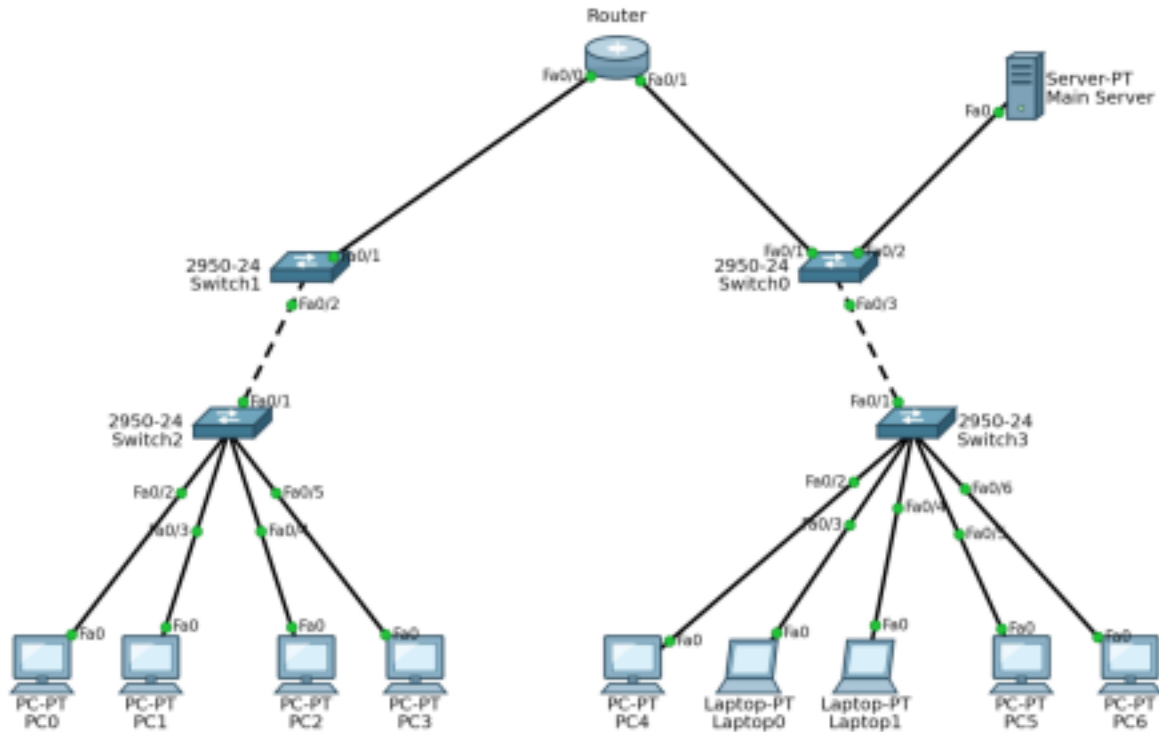


Figure 2: Topology for Task # 02

Note

Assign IP addresses to **both interfaces of the router** using your roll number. The network format should be **XX.XX.YY.0**, where **XX.XX** is derived from your roll number. For example, if your roll number is **3879**, the network should be **38.79.1.0**. For the second interface, **Y** should be incremented (e.g., **38.79.2.0** for the second network).

Task # 02 Screenshots

PASTE YOUR SSH CONFIGURATION, SSH SESSION AND FA0/1 CONFIGURATION SCREENSHOTS HERE

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ip domain-name example.com
R1(config)#username admin privilege 15 secret cisco123
R1(config)#crypto key generate rsa
The name for the keys will be: R1.example.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2
*Mar 1 0:6:43.344: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 15
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#eixt
      ^
% Invalid input detected at '^' marker.

R1(config)#exi
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

The End!

Good luck — remember to upload your PKT files along with this document.